

Nvidia Final Report (Part 1)

Token Lifecycle and Entropy Analysis in LLMs

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Executive Summary

This is the first part of a two-part summary document for Nvidia’s MATH4AI programme. The first part provides a comprehensive comparative analysis of token dynamics across major Large Language Model (LLM) architectures. To facilitate cross-architectural comparison, the report introduces a Unified Token Lifecycle Framework that maps token processing into five distinct stages: Initialization & Conditioning, Sequence & Spatial Mixing, Transformation & Conditional Routing, Depth & Temporal Evolution, and Projection & Emission.

The analysis evaluates four primary architectural families through this lens:

- **Transformers:** Characterized by explicit, all-to-all dense interactions and continuous spatial depth, where geometric forces naturally lead to phenomena like token collapse and “Attention Sinks.”
- **Mamba / State Space Models (SSMs):** Defined by sequential recurrence and input-dependent selectivity, which allow the model to efficiently filter and process tokens through “hidden” attention mechanisms.
- **Diffusion LLMs:** Departs from standard left-to-right processing by utilizing bidirectional context and iterative temporal denoising to simultaneously refine token certainty.
- **Mixture-of-Experts (MoE):** Distinguished by conditional routing, dynamically assigning specialized computational experts based on an individual token’s complexity gradient and semantic category.

Furthermore, the report maps cross-architecture phenomena by comparing metrics such as entropy, intrinsic dimension, representational rank, and circuit complexity. It concludes with a literature synthesis of foundational academic papers, exploring the theoretical limits, geometric vulnerabilities, and emerging hybrid models that define the current frontier of sequence modeling.

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Chapter 1

Token Lifecycle

1.1 Introduction

Unified Token Lifecycle Framework

While there is no “correct” classification of a token lifecycle into stages, I have classified them into five steps for ease of comparison. Stage 1: Encoding Stage 2: Attention (within a block) Stage 3: FFN + nonlinear transformation (within a block) Stage 4: Trajectory across depth Stage 5: Output

1. **Initialization & Conditioning (Encoding):** This stage represents the genesis of the token. It focuses on translating raw discrete or continuous data into a unified latent space, augmented with the necessary structural metadata required for the specific architecture to understand its place in space or time.
 - *Core Function:* Embedding generation, positional encoding, and the injection of global conditional priors (e.g., diffusion timesteps or system prompts).
2. **Sequence & Spatial Mixing (Interaction):** This stage governs how a token gathers context from its environment. It focuses entirely on inter-token communication, defining the mechanism through which a token aggregates external signals from other tokens in the sequence or spatial grid.
 - *Core Function:* Information routing across the sequence, context assimilation, and the updating of a token’s representation based on its neighbors or a compressed hidden state.
3. **Transformation & Conditional Routing (Processing):** Once a token has gathered context from its environment, this stage governs its internal, intra-token refinement. It involves non-linear feature expansion and, crucially, the decision-making process regarding *where* or *how* the token should be processed next based on its current state.
 - *Core Function:* Independent token transformation, feature extraction, and conditional pathway selection (routing/gating) through specialized sub-networks.

4. **Depth & Temporal Evolution (Integration):** This stage models the cumulative trajectory of the token. Rather than looking at a single layer, this stage captures the token’s lifecycle as an ongoing dynamical process, tracking how its representation updates, stabilizes, or denoises as it moves through network depth or iterative temporal steps.
 - *Core Function:* Residual accumulation, state-space transitions over continuous depth, and iterative signal-to-noise refinement.
5. **Projection & Emission (Output):** The final stage of the lifecycle. The matured token representation is decoded from the high-dimensional latent space back into the target observable space, finalizing its journey.
 - *Core Function:* Dimensionality reduction, final layer normalization, and mapping to a probability distribution (e.g., vocabulary logits) or continuous value (e.g., noise estimation or pixel values).

Before any LLMs understand text, text is converted into numbers through a process known as tokenization. While there are many distinct tokenization methods, efficient tokenization methods reduce the computing power needed for training and inference. Additionally, tokens (not bits) are the fundamental unit of semantic information. LLMs act as discrete-time channels, and the semantic information flow (directed information density) naturally progresses and accumulates over the token sequence. By analyzing token behavior across each LLM infrastructure, we can analyze each current LLM infrastructure more deeply.

1.2 Transformer

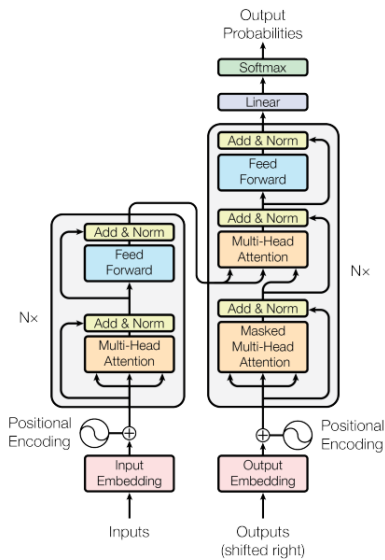


Figure 1: Full Lifecycle of Token in the Transformer Architecture

Fig 1. illustrates a basic transformer architecture. Input is encoded, then passed through the transformers many layers. Most notably, each layer has a multi-head attention and a feed-forward network. At the end of the layer, the final vector is decoded and passed as output (Vaswani et. al., 2017).

1.2.1 Stage 1: Initialization & Conditioning (Encoding)

Not all transformers have encoders. However, since the original transformer infrastructure introduced had an encoder-decoder structure, it will be included in this token lifecycle overview (Vaswani et. al., 2017). The lifecycle begins by mapping discrete inputs into a continuous, high-dimensional space where they can be mathematically manipulated. Raw tokens are converted into continuous vectors via learned **Embeddings**. Embeddings help capture relationship and context. For instance, related terms like “cat” and “dog” will have closer vector distances.

Because the architecture processes tokens in parallel without inherent sequential logic, positional encodings (such as sinusoidal waves or RoPE) are injected to provide order (information on the position of each word). Tokens are initially projected into a high-dimensional embedding space, typically around 10^3 dimensions (Song et al., 2025).

1.2.2 Stage 2: Sequence & Spatial Mixing (Interaction)

This stage governs how a token gathers context from its environment, defining the mechanism through which a token aggregates external signals from other tokens in the sequence.

Explicit Dense Interaction The first major component inside each replicated layer of the transformer is the multi-headed attention (Vaswani et. al., 2017). This is arguably the most influential component when it comes to identifying and incorporating meaningful information about a token’s role in the entire sequence. Multiple heads in this attention mechanism are each specialized in capturing different linguistic aspects simultaneously. The core mechanism is explicitly calculated via self-attention weights, resulting in $\mathcal{O}(n^2)$ computational complexity.

At the heart of every Transformer layer is the scaled dot-product attention mechanism (Vaswani et. al., 2017). For a sequence of tokens, the attention weights are calculated by taking the queries (Q) and keys (K), computing their dot product, and passing them through a softmax function:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V$$

To see what happens to an individual token, we can look at the specific attention weight (α_{ij}) that a token i (the query) assigns to another token j (the key):

$$\alpha_{ij} = \frac{\exp(q_i \cdot k_j / \sqrt{d})}{\sum_{m=1}^N \exp(q_i \cdot k_m / \sqrt{d})}$$

Three Phase Model Historically, we might have assumed that each multi-attention layer does roughly the same thing. However, recent research has shown that tokens go through different processes across different attention layers (Queipo-de-Llano et al., 2026). This can be divided into three phases: Mix, Compress (in Stage 3), and Refine (in Stage 5). These phases will be elaborated in different stages of the Unified Token Lifecycle for transformers.

Mix (Early Layers) However, the model actually gathers all possible context, computing how every word relates to the others in a “messy” initial phase (Queipo-de-Llano et al., 2026). This initial mixing stage is deliberately limited to prevent over-mixing and representational collapse that would occur with extended uniform attention.

Attention Sinks and Geometric Forces During this mixing phase, massive token activations cause the model to direct a disproportionate amount of attention to the very first few tokens in a sequence, even if those initial tokens contain no semantic meaning (Attention Sinks) (Queipo-de-Llano et al., 2026). Furthermore, the softmax attention matrix applies an inescapable, attractive force on the tokens. By averaging values, tokens constrained to a hypersphere rapidly lose their angular dispersion and are driven to collapse into a single mass unless countered (Rigollet, 2026).

1.2.3 Stage 3: Transformation & Conditional Routing (Processing)

Once a token has gathered context, this stage governs its internal, intra-token refinement. Tokens pass through Feed-Forward Networks (FFN) to undergo non-linear feature expansion. The FFN step is applied independently on each token, refining the contextual patterns already integrated to yield useful “knowledge” from them. These layers are also supplemented with residual connections and layer normalizations. As a result, at the end of this step, we have a transformer layer with an updated representation that will be input to the next transformer layer.

Compress (Middle Layers) The model squishes the accumulated information into a dense, high-energy summary, creating a **Compression Valley** (Queipo-de-Llano et al., 2026). During this phase, recent research shows that while effective models have a high-dimensional working space where features are captured, tokens in such models can also be represented in approximately 10-dimensional submanifolds that closely resemble human semantic spaces (Song et al., 2025).

1.2.4 Stage 4: Depth & Temporal Evolution (Integration)

This stage models the cumulative trajectory of the token, tracking how its representation updates, stabilizes, or denoises as it moves through network depth.

The Residual Stream & Information Bottlenecks This stream is the sum of the outputs from all previous layers and the original embedding. It functions as an internal communication channel, allowing the model to preserve a version history across depth (Ali et al., 2025). However, because a token must compress all historical data into its current layer’s residual stream to pass context forward, it suffers from an information bottleneck (Gerasimov et al., 2025). Passing through successive self-attention blocks can cause severe feature homogenization, gradually stripping tokens of their unique semantic features (Kang et al.).

Transformers as Continuous Flows: The ODE Interpretation The discrete, layer-by-layer update of a transformer has a precise mathematical analogue in continuous dynamical systems. Fein-Ashley (2025) formalizes this by showing that each residual layer update—where a token representation x_n is updated as:

$$x_{n+1} = x_n + \frac{1}{N}f(x_n)$$

is structurally identical to a **forward Euler step**, the simplest numerical method for solving ordinary differential equations. Here, $f(x) = g(x) + h(x)$ decomposes into the self-attention sublayer g and the feedforward sublayer h , and N is the total number of layers.

As N grows, the **Transformer Flow Approximation Theorem** establishes that token trajectories converge uniformly at rate $\mathcal{O}(1/N)$ to the unique solution of the continuous

ODE:

$$\frac{dx(t)}{dt} = f(x(t)), \quad x(0) = E(\text{input})$$

A key consequence emerges when the mapping f satisfies a **one-sided Lipschitz condition** with a negative constant λ :

$$\langle f(x) - f(y), x - y \rangle \leq \lambda \|x - y\|^2, \quad \lambda < 0$$

Under this condition the dynamics are contractive: any perturbation δ_0 between two token representations decays exponentially with depth:

$$\|\delta(t)\| \leq \|\delta_0\| \cdot e^{\lambda t}$$

This is the formal basis for the empirical observation that deep transformers are robust to noise—small input variations are smoothed out as representations propagate through layers. However, contractivity is indiscriminate. The same force that suppresses noise also suppresses semantic distinctions. When λ is strongly negative, diverse token representations are pulled together regardless of meaning, risking **representation collapse** (Fein-Ashley, 2025).

Note: Fein-Ashley (2025) was withdrawn by its author in May 2025. The theoretical framework is consistent with the broader Neural ODE literature, but the specific proofs should be treated with appropriate caution pending clarification of the withdrawal.

1.2.5 Stage 5: Projection & Emission (Output)

The final stage ends the token’s internal journey. The matured token representation is decoded from the high-dimensional latent space back into the target observable space.

Refine (Final Layers) The model shifts from macro-context to micro-prediction, polishing the dense summary to focus purely on the next specific word (Queipo-de-Llano et al., 2026). From a token-centric perspective, the uniformity breaks. The tokens stop sharing that identical global summary and diverge rapidly. Each token’s vector transforms into a highly specific, individualized pointer meant to trigger the exact vocabulary prediction for whatever word comes next.

Output The final token representations undergo a Final LayerNorm, followed by a Linear projection and a Softmax function (Vaswani et. al., 2017). During this final projection, models utilize distinct, family-specific decision strategies that balance expansion (considering different possibilities) and pruning (removing unlikely words) to finalize the prediction (Ali et al., 2025). This final mechanical process generates a probability distribution over the entire vocabulary to predict the next token autoregressively, completing the token lifecycle.

1.3 The Mamba Token Lifecycle Framework

Why SSMs and Mamba While Transformers rely on attention mechanisms, SSMs (like Mamba) act like upgraded Recurrent Neural Networks. SSMs excel in linear inference speeds ($\mathcal{O}(n)$ speed relative to transformer $\mathcal{O}(n^2)$) and low memory usage, but lack the robust understanding of context that transformers possess, making it a poor option for complex reasoning models like LLMs. This creates a tradeoff between speed and accuracy for LLMs.

Mamba, as an evolution of SSMs, is the first alternative to the transformer architecture while having the speed of SSMs. It does so through two innovations: the first is the selective state space model, which provides Mamba with a crucial capability previously possessed only by transformer models: the ability to selectively focus on or ignore specific parts of past input history based on their present relevance. The other is hardware-aware parallel scan, an algorithm that optimizes the way a graphics processing unit (GPU) handles the model’s computations in its memory hierarchy to maximize speed and computational efficiency.

The Mamba architecture fundamentally reimagines how tokens flow through a sequence. While Transformers rely on explicit, all-to-all geometric forces, Mamba operates as a continuous dynamical system optimized for hardware efficiency. However, beneath this rigid sequential pipeline lies a massive, distributed ensemble of hidden, dynamic token interactions.

1.3.1 Stage 1: Initialization & Conditioning (Encoding)

This stage sets the foundation, transforming discrete text into a mathematical object capable of being manipulated within a continuous state space.

Linear Projection & Branching After raw tokens are converted into continuous vector embeddings, the input vector undergoes a linear projection that expands its dimensionality. Crucially, Mamba splits this expanded token into two distinct parallel computational paths: one destined for the Selective State Space Model (SSM) core (x_{proj}), and the other for a parallel multiplicative gating mechanism (z_{proj}) (Gu & Dao, 2023).

Inherent Temporal Grounding Unlike Transformers, Mamba does not require explicit positional encodings (like sinusoidal waves or RoPE) to be injected into the token. Because its foundation relies on continuous-time differential equations, the sequence and temporal order are inherently baked into the mathematics of the state space.

1.3.2 Stage 2: Sequence & Spatial Mixing (Interaction)

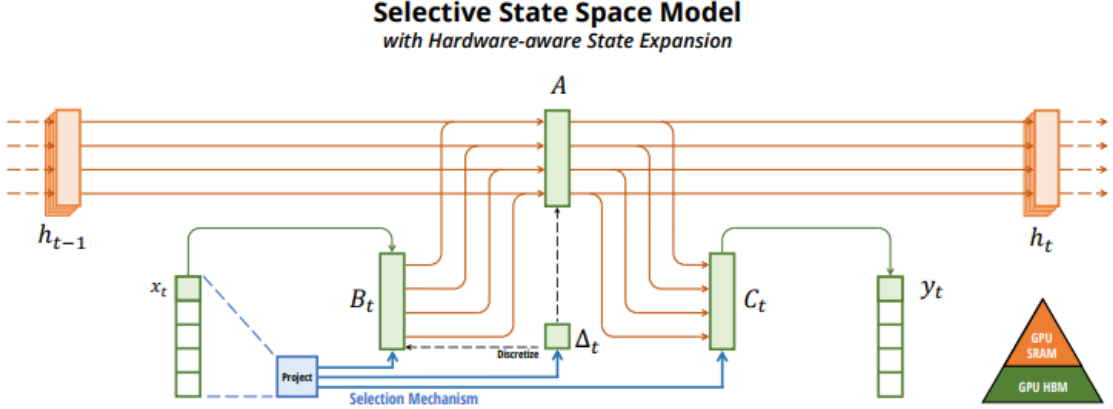


Figure 2: Selective SSM architecture

This stage dictates how a token gathers context from its environment. On the surface, Mamba appears to destroy standard token interaction, but mathematically, it recreates it efficiently through the State Space Model(SSM) recurrence.

$$h'(t) = Ah(t) + Bx(t) \quad (1.1)$$

To process discrete tokens in a sequence, this continuous system is discretized using a timescale parameter Δ , transforming it into the foundational SSM recurrence:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t \quad (1.2)$$

$$y_t = Ch_t \quad (1.3)$$

Here, x_t is the current token input, h_t is the hidden state (the accumulated context), and y_t is the output. The matrices $\bar{A}$ and $\bar{B}$ are the discretized versions of the continuous parameters A and B . Evidently, there is a radical mechanistic difference in how these architectures process tokens. Transformers use all-to-all dense interactions, while Mamba relies on sequential, state-mediated updates (Chen et al., 2024). Tokens are processed sequentially and squashed into a compressed 1D hidden state, seemingly destroying the direct token-to-token maps seen in Transformers (Pitorro & Treviso, 2025).

Hidden Attention Generation Recent studies have suggested that Mamba does not merely approximate attention, but rather generate “hidden” attention matrices on the fly (Ali et al., 2025). By unrolling this sequential process mathematically, it is proven that Mamba implicitly forms highly structured dependencies over time, dynamically prioritizing both recent and highly salient distant tokens without ever actually computing a full $\mathcal{O}(n^2)$ matrix (Pitorro & Treviso, 2025).

1.3.3 Stage 3: Transformation & Conditional Routing (Processing)

Once context is flowing into the hidden state, this stage covers the non-linear transformations and the model’s core decision-making process.

Local Convolutional Extraction Before entering the SSM, the token representation is fed through a 1D convolution layer (Gu & Dao, 2023). This acts as a localized filter, extracting immediate syntactic patterns between neighboring tokens, which frees the deeper state space to focus entirely on global dependencies. The output of the convolution layer serves as the input to a nonlinear activation function. For instance, the original Mamba infrastructure uses Sigmoid Linear Unit (SiLU). The separate gating mechanism path, z_{proj} , also goes through a non-linear activation function.

Input-Dependent Selectivity The defining feature of Mamba is its input-dependent gating—unlike older fixed-state models, Mamba decides what to remember based on what it is currently reading (Gu & Dao, 2023). The mathematical parameters defining the state space transition are continuously recalculated based on the specific token at that exact step.

1.3.4 Stage 4: Depth & Temporal Evolution (Integration)

This stage models the cumulative trajectory of the token as it updates and stabilizes across network depth. This is where Mamba’s “Selective” mechanism mathematically grounds the model’s ability to retain context and filter noise.

Input-Dependent Selectivity & Counteracting Decay Standard recurrent networks and vanilla SSMs suffer from exponential memory decay because their transition matrices (A , B , C , and Δ) are static—they apply the exact same mathematical operation regardless of the input (Huang et al., 2025). Mamba breaks this limitation and achieves “selectivity” by making the step size (Δ), B , and C explicitly dependent on the current token input x_t :

$$B_t = \text{Linear}(x_t)$$

$$C_t = \text{Linear}(x_t)$$

$$\Delta_t = \text{Softplus}(\text{Parameter} + \text{Linear}(x_t))$$

Because $\bar{A}$ and $\bar{B}$ are derived using the now-dynamic Δ_t , the entire recurrence equation becomes fundamentally input-dependent. By dynamically adjusting these parameters on a per-token basis, Mamba actively counteracts that natural exponential decay (Huang et al., 2025). If a token is irrelevant, the model can mathematically “shrink” Δ_t to ignore it; if a token is critical, it can expand it to heavily update the hidden state.

Multi-Scale Filtering Because the math is dynamic, the token acts as a sophisticated, multi-scale filter. It deliberately holds onto specific, highly relevant historical tokens for massive context windows, while aggressively forgetting useless intermediary tokens (Huang et al., 2025).

Circuit Complexity Equivalence Despite processing tokens one by one through a constrained state space, Mamba suffers no fundamental theoretical disadvantage in expressivity; it is mathematically capable of computing the exact same class of functions over a token sequence as a full self-attention mechanism (Chen et al., 2024).

1.3.5 Stage 5: Projection & Emission (Output)

The final stage ends the token’s internal journey, aggressively shifting from continuous latent states back out to discrete predictive probabilities.

Gated Recombination The output emerging from the sequential SSM path is multiplied element-wise by the activation from the parallel gating path established back in Stage 1. This gate acts as a volume dial—suppressing the contextual update if irrelevant, or amplifying it if highly semantic (Gu & Dao, 2023).

Residual Accumulation The resulting recombined vector is projected back down to its original dimension and added directly to the residual stream.

Emission The final token representation undergoes normalization and a linear projection out to the vocabulary size, generating a probability distribution to autoregressively predict the next word.

1.4 The Diffusion Token Lifecycle Framework

Unlike autoregressive models that accumulate meaning sequentially from left to right, Diffusion LLMs represent a radical departure in sequence processing. In this architecture, tokens are generated through an iterative temporal process of continuous refinement rather than step-by-step spatial accumulation (Nie et al., 2025).

Three limitations of transformer architecture that diffusion models attempt to fix are error propagation (error in earlier token leading to error in latter tokens), indirect control, and slow token-by-token generation (model needs to predict earlier token before subsequent). In contrast, diffusion models offer iterative refinement, greater control, and faster sampling.

Additionally, diffusion models can be classified into discrete masked diffusion and continuous diffusion. As most of the literature reviewed (LLaDA, MDLM) are of the discrete kind, this framework will **focus on discrete masked diffusion models** (Nie et al., 2025; Sahoo et al., 2024).

1.4.1 Stage 1: Initialization & Conditioning (Encoding)

This stage represents the genesis of the token, translating raw data into a latent state. However, in diffusion models, tokens completely break the standard autoregressive lifecycle.

Forward Masking (Intentional Corruption) During training, tokens are processed through a forward masking process where they are intentionally corrupted by noise.

The Masking Formula In discrete masked diffusion models (like MDLM and LLaDA), a clean token $x^{(\ell)}$ at position ℓ transitions to a special mask token m over a continuous time variable $t \in [0, 1]$ (Sahoo et al., 2024). This is mathematically defined by the categorical distribution:

$$q_{t|0}(z^{(\ell)} | x^{(\ell)}) = \text{Cat}(\alpha_t \mathbf{e}_{x^{(\ell)}} + (1 - \alpha_t) \mathbf{e}_m)$$

where $\mathbf{e}_v$ is the one-hot vector for a token, and α_t represents the noise schedule.

Inference Genesis During inference, the generation process starts from a sequence of pure noise (e.g., fully masked text), meaning the initial tokens lack any semantic identity prior to the first denoising step (Nie et al., 2025).

1.4.2 Stage 2: Sequence & Spatial Mixing (Interaction)

This stage governs how a token gathers context. Diffusion architectures bypass the restrictive left-to-right attention masks of traditional models.

During inference, all tokens are updated simultaneously. Rather than looking only at past tokens, the model continuously refines its confidence across the entire sequence bidirectionally. It updates the probability distribution of each individual token based on the gradually emerging global context (Nie et al., 2025).

Error Vulnerability In models that unmask tokens progressively rather than simultaneously, if a token is incorrectly predicted early on, that error can severely corrupt the conditioning context for all surrounding tokens.

1.4.3 Stage 3: Transformation & Conditional Routing (Processing)

This stage governs intra-token refinement. In discrete masked diffusion, this translates to the parameterized reverse generation (denoising) process, where the model predicts the true identity of masked tokens.

Markov Transition Dynamics (The Absorbing State) In discrete language diffusion, token corruption and generation are defined by a Markov transition matrix. Under a strict masked diffusion paradigm, the ‘[MASK]’ token acts as an “absorbing state”. This means during the forward corruption process, once a semantic token transitions into a mask, it loses all identity. The reverse process (generation) is the iterative effort to pull the token out of this absorbing state and back into the semantic vocabulary space.

The Transformer Backbone Crucially, the denoising network x_θ is parameterized by a standard Transformer architecture. This means the entire spatial sequence-mixing lifecycle described in Section 1 (dense attention, feature extraction, and residual flow) is actually embedded *inside* each individual temporal denoising step of the diffusion model.

Iterative Semantic Locking Rather than predicting the entire sequence in a single forward pass, the model iteratively predicts the clean sequence from the partially masked sequence. At each timestep, tokens with the highest prediction confidence are “locked in” (un-masked), which sequentially narrows the contextual possibilities for the remaining masked tokens (Shi et al., 2025).

Reverse Prediction Matrix The denoising network x_θ takes a partially masked sequence z_t and outputs a token probability matrix predicting the clean tokens:

$$\mathbf{P} = \text{softmax}(x_\theta(z_t))$$

This provides the mathematical basis for substituting the unknown “[MASK]” tokens with model predictions, generating the specific cross-entropy losses needed for the next temporal step.

1.4.4 Stage 4: Depth & Temporal Evolution (Integration)

Instead of just tracking a token across spatial network depth, this stage tracks tokens across temporal denoising timesteps. Because diffusion models involve entirely different dynamics during optimization versus generation, this stage evaluates both the training and inference trajectories.

Training Trajectory: ELBO Optimization & Semantic Accumulation Diffusion models learn by reversing a noise-corruption process, which can involve high-variance estimates. By using Rao-Blackwellization, this objective is reformulated, reducing variance by conditioning on a subset of variables (Sahoo et al., 2024). This technique rewrites the complex diffusion objective into a mixture of standard masked cross-entropy losses, yielding a provably tighter Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}}(\theta) = \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{z_t} \left[w_t \sum_{\ell \in \mathcal{M}(z_t)} -\log p_\theta(x^{(\ell)} \mid z_t) \right]$$

As tokens undergo this reverse process, their semantic certainty increases continuously. This effectively proves that masked diffusion is mathematically homologous to a continuous, iterative version of Masked Language Modeling (Sahoo et al., 2024).

Inference Trajectory: Dynamic Particle Refinement During inference, a major vulnerability is the accumulation of conditioning errors: an incorrect prediction early in the reverse-denoising path severely corrupts the context for all remaining tokens. To fix this, advanced inference methods model the sequence generation as a dynamic state-space trajectory (Dang et. al., 2025). By applying Particle Gibbs sampling, the model runs multiple parallel token trajectories (“particles”) and dynamically resamples them based on their likelihood scores. This allows tokens to mathematically “change their minds” mid-generation if their current trajectory leads to low-probability semantic states, actively healing context errors before finalizing the output.

1.4.5 Stage 5: Projection & Emission (Output)

The final stage finalizes the token’s journey, mapping the mature representation to the target observable space.

Final Denoising Step As the diffusion timetable t approaches 0, the final noise is stripped away. The fully unmasked sequence reaches peak semantic certainty.

Simultaneous Emission Unlike autoregressive models, the probability distributions for the entire vocabulary are mapped simultaneously across all token positions (Nie et al., 2025). The generation stops, and the text sequence is emitted as a cohesive, bidirectionally informed whole.

1.5 The Mixture-of-Experts (MoE) Token Lifecycle

Unlike standard dense models where every token undergoes the exact same mathematical operations, the MoE lifecycle is defined by conditional routing. In this framework, tokens are triaged based on their complexity, semantic category, and temporal confidence, meaning two tokens in the same sequence may experience entirely different lifecycles.

1.5.1 Stage 1: Initialization & Conditioning (Encoding)

This stage represents the genesis of the token. MoE models generally inherit their initialization mechanics directly from their backbone architecture (e.g., autoregressive or diffusion).

The Universal Baseline Tokens are converted from discrete text into continuous, high-dimensional vector embeddings, augmented with positional metadata. At this stage, all tokens are treated equally. They share the same geometric space before arriving at the deeper layers where conditional routing begins.

1.5.2 Stage 2: Sequence & Spatial Mixing (Interaction)

Before a token can be efficiently routed to an expert, it must gather contextual information about its environment.

The Contextual Prerequisite In most MoE models, token interaction is handled by standard self-attention mechanisms before the token reaches the MoE feed-forward layers. Tokens gather global context via $\mathcal{O}(n^2)$ attention matrices.

Hybrid Interaction (The Alternating Rhythm) In advanced hybrid MoE models (such as Jamba), a token experiences a radically shifting environment during its mixing stage. Instead of passing through a uniform sequence of attention matrices, the token is subjected to an alternating computational rhythm (Lieber et al., 2024). When passing through Mamba (SSM) blocks, the token’s experience is constrained and sequential: it acts as a synthesizer, compressing its historical context into a fixed-size, rolling hidden state to efficiently filter out background noise. However, when the token periodically enters an interleaved Transformer block, it is momentarily pulled out of this compressed flow and projected into a dense, global attention matrix. In this phase, the token breaks free from its sequential constraints to execute sharp, precise memory retrievals, establishing direct mathematical links to specific past tokens. For the token, this creates a lifecycle that oscillates between building a broad, generalized summary of the past and performing high-precision associative recall, all before it reaches the MoE routing gates (Lieber et al., 2024). More recent frontier models extend this alternation further by routing tokens through a continuous latent space that dynamically arbitrates between sequential Mamba processing and dense self-attention based on each token’s specific memory demands (Chandiramani et al., 2026).

1.5.3 Stage 3: Transformation & Conditional Routing (Processing)

This is the defining stage of the MoE lifecycle. When a token reaches an MoE layer, it faces a series of distinct mathematical bottlenecks, which different architectures solve through highly specialized routing functions.

The standard mathematical formulation for an MoE layer applied to a token x is:

$$y = \sum_{i=1}^N g_i(x) E_i(x)$$

Where N is the number of experts, $E_i(x)$ is the output of the i -th expert, and $g_i(x)$ is the output of the gating network, typically governed by a Top- K routing function:

$$G(x) = \text{softmax}(\text{TopK}(x \cdot W_g, k))$$

However, there have been multiple proposed models on optimizing the process of the selection of experts to improve model results. Each architecture attempts to tackle a different limitation of the Top- K routing process.

Problem 1: Load Balancing & Compute Allocation Standard MoE architectures force a rigid Top- K routing policy (e.g., routing every token to exactly 2 experts), which mechanically assumes every token requires the exact same amount of computational work (Gülmez, 2026). **The Solution (DynaMoE):** DynaMoE proves tokens inherently contain a measurable “complexity gradient” (Gülmez, 2026). It dynamically allocates experts based on a computed confidence threshold. Simple syntactic tokens (like “the”) might bypass the MoE layer entirely (0 experts), while highly complex semantic tokens trigger 4 or 5 experts simultaneously, explicitly linking a token’s information entropy to its computational routing requirement (Gülmez, 2026).

Problem 2: Expert Collapse & Routing Geometry Standard Top- K softmax routing occurs in a flat Euclidean space, which frequently leads to “expert collapse”—a failure mode where all tokens blindly flood the same few experts while ignoring others (Shihab, Akter, & Sharma, 2026). **The Solution (Grassmannian MoE):** This architecture reformulates routing as a purely geometric problem, mapping tokens and expert centroids onto a curved Grassmann manifold (Shihab et al., 2026). Instead of softmax, tokens are routed based on the Bingham distribution. In this curved geometry, experts naturally repel each other into orthogonal subspaces, preserving routing entropy and mathematically preventing expert collapse (Shihab et al., 2026).

Problem 3: Gradient Conflicts In dense models, forcing vastly different semantic tokens through the same parameters causes geometric “gradient conflicts,” where optimizing the weights for one token type degrades the model’s understanding of another (Liang et al., 2024). **The Solution (Mixture-of-Transformers):** This model routes tokens to specialized Transformer experts to partition the global loss landscape into distinct, strongly convex subtasks (Liang et al., 2024). This ensures gradients from different semantic clusters remain orthogonal. Recent theoretical analyses of this framework prove that this specialization transforms the optimization problem, enabling an exponentially fast $\mathcal{O}(\log(\epsilon^{-1}))$ convergence rate for specialized subsets (Chen et al., 2025).

1.5.4 Stage 4: Depth & Temporal Evolution (Integration)

After tokens are processed by their respective experts, their individual trajectories must be reintegrated. This evolution occurs across two fundamentally different axes depending on the backbone architecture.

Spatial Trajectory (Autoregressive MoEs) In models like DynaMoE (Gülmez, 2026) or MoT (Liang et al., 2024), tokens evolve across network depth. The model must continuously stabilize these highly varied token vectors to ensure downstream attention heads can still read the global sequence context.

Temporal Trajectory (DiffMoE) In Diffusion MoEs, tokens evolve across iterative denoising timesteps rather than spatial layers. In standard diffusion, tokens are processed uniformly at every timestep, which is mathematically wasteful because some tokens resolve

faster than others (Shi et al., 2025). **DiffMoE** introduces dynamic, token-level temporal routing. As tokens achieve high semantic certainty early in the reverse diffusion process, their computational requirement decays; they are dynamically routed to “lighter” experts or dropped entirely, while highly uncertain tokens are continually routed to high-capacity experts (Shi et al., 2025).

1.5.5 Stage 5: Projection & Emission (Output)

The final stage concludes the token’s highly individualized journey.

Recombination and Decoding Regardless of how many experts a token utilized, or whether it skipped layers entirely, its final continuous vector is deposited back into the residual stream. It undergoes final layer normalization to stabilize its variance.

Vocabulary Expansion The latent vector is then linearly projected to project its dimensions out to the full size of the vocabulary space. A final Softmax operation converts these raw logits into a probability distribution, allowing the token to either be autoregressively emitted or temporally locked in as a finalized prediction.

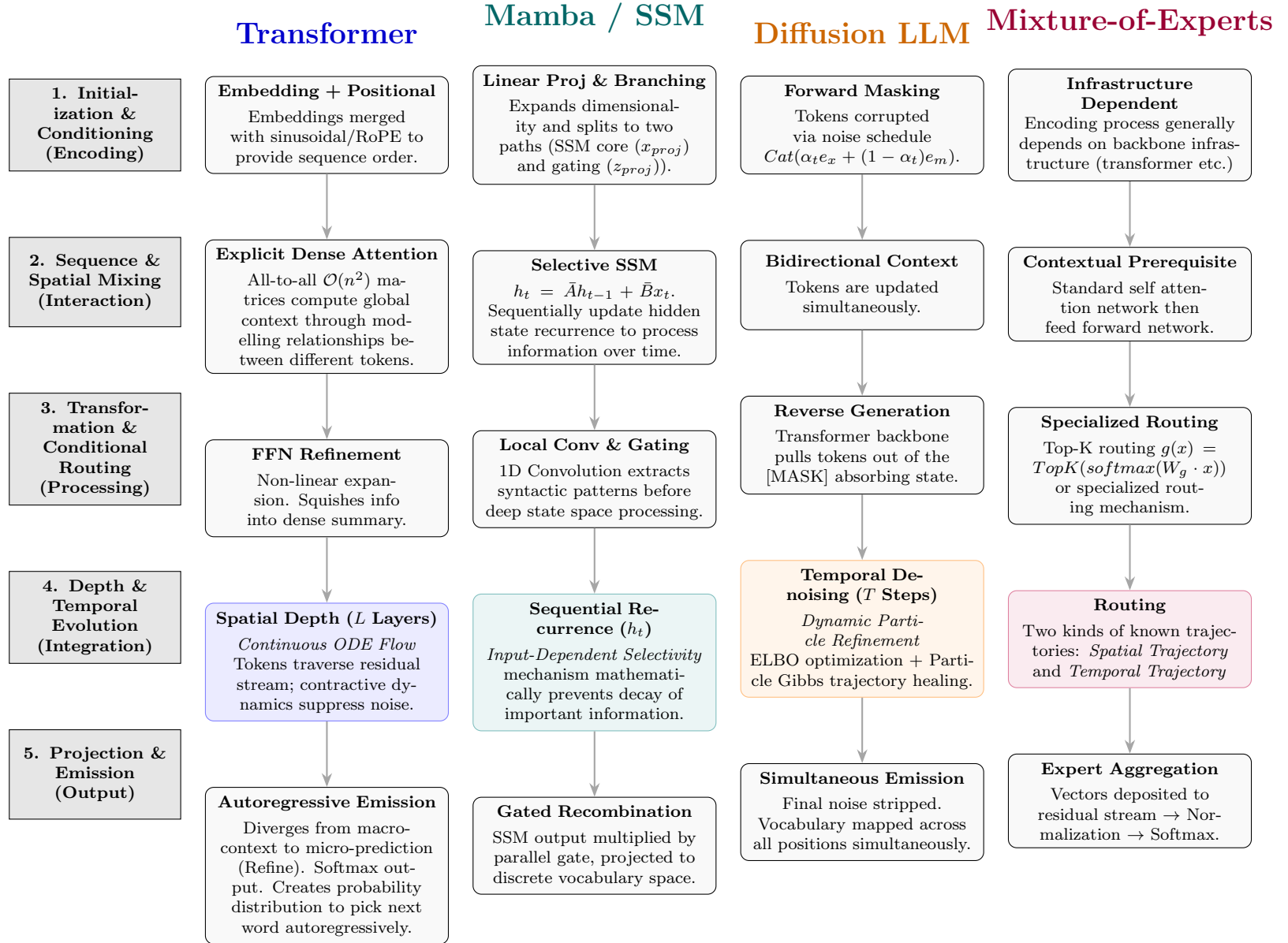


Figure 3: The 5-Stage Spatiotemporal Token Lifecycle Framework. Stage 4 highlights the core geometric divergence: spatial depth (Transformers/MoE), sequential recurrence (Mamba), and iterative temporal refinement (Diffusion).

Chapter 2

Cross Architecture Phenomenon Mapping

Note: Token collapse is about the data points (tokens) merging together, while dimensional reduction is about the underlying space (features) shrinking.

Note: I have made a distinction between Computational Complexity and Circuit Complexity. Previous sources have used "computational complexity" to address the minimum size and depth of physical or abstract logic gates required to compute a specific function. In this segment, this idea will be termed as circuit complexity. On the other hand, computational complexity will refer to time complexity.

Token clustering / collapse

Transformer	Well-studied: Token collapse is an inherent, energy-minimizing pathology of self-attention. Mathematically, attention acts as a Wasserstein gradient flow on a hypersphere, naturally dragging all tokens into a uniform Dirac mass (Rigollet, 2025). Before total collapse, however, tokens form expressive semantic clusters (Rigollet, 2025). Transformers preserve these clusters by utilizing layer normalization as a "speed regulator" (Karagodin et al., 2025). While Post-LN maintains a constant velocity that accelerates collapse, Pre-LN and Peri-LN configurations allow token magnitudes to grow (Karagodin et al., 2025). This growth dynamically decays particle velocity, effectively braking the system and freezing tokens inside their useful multi-clustered states to maintain representation rank (Karagodin et al., 2025).
Mamba / SSM	Unknown: It remains unknown if Mamba's sequential processing creates an analogous mathematical collapse.

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Diffusion LLM	Unknown: It is currently unknown whether tokens cluster or collapse into submanifolds during denoising steps. While there is existing literature on diffusion models, it cannot be safely generalised for diffusion LLMs.
MoE	Partially studied (Expert Collapse): <i>Grassmannian Mixture-of-Experts</i> proves that standard Euclidean routing causes tokens to flood identical experts, but mapping to a Grassmannian manifold mathematically prevents this by forcing experts into orthogonal subspaces.

Dimensional reduction

Transformer	Well-studied: <i>Bridging the Dimensional Chasm</i> shows a geometric framework where tokens compress from high-dimensional embedding spaces (10^3) onto ≈ 10 -dimensional submanifolds representing human semantics (Song et al. 2025). Such phenomenon is evident in effective LLM models.
Mamba / SSM	Partially Studied: Mamba compresses past tokens into a fixed-size recurrent hidden state matrix. When sequence lengths vastly exceed the model’s training threshold, the model encounters a memory capacity ceiling (Chen et al., 2024).
Diffusion LLM	Partially Studied: In deep Diffusion Transformers, researchers have identified a structural vulnerability called a mean-dominated collapse state, triggered by a phenomenon known as Mean Mode Screaming (MMS). When scaling DiTs (Diffusion Transformer) to extreme depths (400 to 1000 layers) for video or high-fidelity image generation, the residual stream becomes progressively dominated by the pure-mean component. The centered variation component gets mathematically squeezed out ($\ c(X)\ _F \rightarrow 0$). The tokens experience a severe dimensional reduction, homogenizing into a uniform, low-variance state that completely suppresses visual variation and crashes generation quality.
MoE	Unknown: It remains an open gap how dimensional reduction impacts routing efficiency or if routed tokens change their intrinsic dimension.

Information compression

Continued on next page

Transformer	3-Phase Pattern: Llano et al. (2025) outlines a "Mix-Compress-Refine" pattern where intermediate layers squish information into a dense, high-energy summary. Such observations attempt to explain how transformer LLMs organize their computation in depth by controlling attention and representational compression via massive activations.
Mamba / SSM	Multi-Scale Filtering: <i>Understanding Input Selectivity in Mamba</i> models Mamba as actively counteracting memory decay by deliberately holding onto critical tokens and aggressively forgetting useless ones (Huang et al., 2025).
Diffusion LLM	Iterative Semantic Locking: An example of diffusion LLM, <i>Simple and Effective Masked Diffusion Language Models</i> , proves continuous semantic certainty increases through reverse prediction, yielding a tighter ELBO (Sahoo, 2024).
MoE	Complexity Gradients: <i>DynaMoE</i> links token information entropy to compute, dropping simple tokens (0 experts) while heavily routing complex tokens (4–5 experts) (Gulmez, 2026).

Representational rank

Transformer	Rank decreases with depth: Without non-linearities (like MLPs) or skip-connections, token representations rapidly homogenize across successive layers. The token matrix $X \in \mathbb{R}^{n \times d}$ experiences complete rank collapse, where all singular values except the first one drop to zero ($\sigma_2(X) \rightarrow 0$). The tokens collapse into a uniform, rank-1 subspace (Dong et al., 2021).
Mamba / SSM	Similarity to transformer: According to Yang et al. (2025), Mamba’s token-mixing framework acts as a structural low-rank approximation of traditional Softmax Attention. Unlike Transformers, which possess the geometric freedom to construct full-rank token matrices, Mamba’s linear recurrence inherently restricts its feature space to a lower matrix rank. This structural bottleneck means Mamba has a weaker absolute representational capacity than standard self-attention. However, it scales linearly and retains a significantly higher matrix rank and expressivity than pure Linear Attention, positioning Mamba as a theoretically optimal, high-efficiency middle ground for sequence modeling.
Diffusion LLM	Unknown: Literature does not explicitly cover representational rank limits in diffusion structures.
MoE	Unknown: It is unknown how rank evolves with depth in routed models.

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Circuit complexity

Transformer	TC⁰ Class: Can compute polynomial precision functions via $\mathcal{O}(n^2)$ interactions.
Mamba / SSM	TC⁰ Class: <i>The Computational Limits of State-Space Models</i> proves Mamba is in the exact same TC^0 complexity class as Transformers, matching their expressivity despite sequential processing (Chen, 2024).
Diffusion LLM	Unknown: Formal computational bounds for discrete language diffusion generation are not covered.
MoE	Varies: Complexity depends inherently on the gating function and expert assignment structure.

Computational Complexity

Transformer	$\mathcal{O}(T^2d^2)$ training, but dominates to $\mathcal{O}(T^2)$ as sequence length grows. Inference: $\mathcal{O}(Td + d^2)$.
Mamba / SSM	Computation and memory scale linearly with sequence length during training. During inference, unrolling the model autoregressively requires only constant time per step, since no cache of previous elements is needed (Gu & Dao, 2023).
Diffusion LLM	No standard training time complexity. For a Transformer-backed diffusion model, each reverse denoising step executes a full bidirectional attention pass at $\mathcal{O}(T^2d)$ cost (Sahoo et al., 2024), and this is repeated across K iterative sampling steps, so total generation cost scales with both the number of denoising steps and sequence length (Nie et al., 2025).
MoE	$\mathcal{O}(T^2d)$ training. Inference time complexity of a Mixture of Experts (MoE) layer is practically identical to a dense layer of equivalent active parameters: $\mathcal{O}(k \cdot d_{model} \cdot d_{ffn})$, where k is the number of experts activated per token (typically 2), d_{model} is the hidden dimension, and d_{ffn} is the expert’s inner layer dimension (Shazeer et al., 2017; Fedus et al., 2022).

Implicit vs. Explicit Interaction

Transformer	Explicit Attention: Uses explicit all-to-all dense interactions per layer ($\mathcal{O}(n^2)$).
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Mamba / SSM	Implicit Micro-Routers: <i>The Hidden Attention of Mamba Models</i> shows Mamba dynamically generates implicit attention matrices on the fly, using 3 orders of magnitude more "hidden" micro-routers than Transformer heads (Ali, 2024).
Diffusion LLM	Bidirectional Context: <i>Large Language Diffusion Models</i> shows all tokens update simultaneously during inference, continuously refining confidence globally without left-to-right masks.
MoE	MoEs natively execute explicit self-attention at the sequence layer level to capture token-to-token contextual relationships, leaving this mechanism unchanged from the standard Transformer backbone; only the feed-forward sublayer is replaced with a set of routed experts (Fedus et al., 2022; Lepikhin et al., 2020). This sparse routing mechanism introduces a macro-level, content-dependent interaction pathway between tokens and parameters, distinct from the sequence-level attention graph.

Variable Key: T = Sequence Length, d = Model Hidden Dimension, N = SSM State Dimension, K = Denoising Steps, E = Total Experts, k = Active Experts per token.

Literature Synthesis: Agreements and Contradictions

- **Agreement on Transformer Collapse:** Multiple distinct analytical lenses converge on the same vulnerability. *The Mean-Field Dynamics of Transformers* (Rigollet, 2026) uses geometric/manifold analysis to prove collapse on $\mathbb{S}^{d-1}$, while *Flowing Through Layers* (Fein-Ashley, 2025) reaches the exact same conclusion using continuous dynamical systems (ODEs) proving contractive pull.
- **Agreement on Cross-Architecture Expressivity:** *The Computational Limits of State-Space Models* (Chen et al., 2024) formally bridges Transformers and Mamba by proving they occupy the exact same complexity class (TC^0). Furthermore, *The Hidden Attention of Mamba Models* (Ali et al., 2025) supports this equivalence mechanically, proving Mamba executes a highly factored, implicit form of self-attention.
- **Nuance/Contradiction on the Cause of Representation Collapse:** *Attention Sinks and Compression Valleys* (Queipo-de-Llano et al., 2026) explicitly attributes token representation collapse to massive token activations in the residual stream interacting with LayerNorm. In contrast, *The Mean-Field Dynamics of Transformers* (Rigollet, 2026) argues that collapse is an intrinsic mathematical destiny caused strictly by the softmax attention matrix averaging mechanism, existing independently of specific architectural quirks like residual streams or massive activations.

Chapter 3

Cross Architecture Phenomenon Mapping (II)

3.1 Introduction

This part is similar to the previous part on Cross Architecture Phenomenon Mapping. The key difference is that the previous section explores more about phenomena (Does XXX happen?), while this section is more about metrics and methodology (How exactly did XXX happen, what were the numbers, and were they a fair comparison?).

3.2 Cross-Architecture Metric Compilation

3.2.1 Key Terms

- **Entropy:** The extent of randomness or disorderly in a system. The higher the entropy, the more random the system becomes. Entropy can be modeled using the formula $-\sum p(x) * \log_2(p(x))$ where $p(x)$ is the probability distribution of x .
- **Intrinsic Dimension:** The minimum number of variables that needs to be kept to represent a dataset or loss space without losing significant information.
- **Effective Rank:** (Similar to Intrinsic Dimension) A quantitative, often entropy-based measure that characterizes the true intrinsic dimensionality or “complexity” of matrices, operators, distributions, or feature sets encountered. Intrinsic dimension measures the geometric complexity of data (the degrees of freedom required to describe a dataset), while effective rank measures the linear complexity of a system (the number of principal components needed to capture a feature matrix’s variance).
- **Cosine Similarity:** Measures how similar two datapoints are to each other based on their direction.

Entropy (Information Content)

- **Transformer:** Transformers do not monotonically decrease in uncertainty. Transformers balance depth-invariant expansion (where entropy temporarily rises) and pruning (where entropy drops as the model actively narrows down its decision). The shape of the entropy profile is dependent on the model (Ali et. al., 2025).
- **Mamba / SSM:** No known research has directly measured change in entropy layer by layer. However, we know that semantic information flows and accumulates naturally over the token sequence (Bai, 2025).
- **Diffusion LLM:** No known research has directly measured change in entropy layer by layer. However, we know continuous semantic certainty increases through reverse prediction (Sahoo et. al., 2024).
- **Mixture-of-Experts (MoE):** No unified theory.
- **Comparison Status:** Single-architecture mostly. Bai (2025) offers a unified semantic framework, but layer-wise entropy profiles are only mapped for Transformers.

Intrinsic Dimension (ID)

- **Transformer:** Compresses from high-dimensional embedding spaces (10^3) down to ≈ 10 -dimensional submanifolds in intermediate layers (Song et. al., 2025).
- **Mamba / SSM:** Unknown. Current ID tools rely on explicit layer-by-layer structures, making it difficult to measure in state spaces.
- **Diffusion LLM:** Unknown.
- **Mixture-of-Experts (MoE):** Unknown. Though routing relies on curved mathematical spaces (Grassmannian manifolds), the intrinsic dimension of the routed tokens is unmeasured (Shihab et. al., 2026).
- **Comparison Status:** Single-architecture. Dimensional chasm theory is currently isolated to Transformers.

Effective Rank

- **Transformer:** Rank mathematically decreases with depth as massive token activations in the residual stream push tokens into compression valleys (Queipo-de-Llano et. al., 2026).
- **Mamba / SSM:** Unknown. It is an open question how representational rank evolves with depth in sequential models (Queipo-de-Llano et. al., 2026).
- **Diffusion LLM:** Unknown.
- **Mixture-of-Experts (MoE):** Unknown.

- **Comparison Status:** Single-architecture. Rank degradation is tightly coupled to pre-norm Transformer mechanics.

Token Interaction

- **Transformer:** Traditional transformers use unbounded dot-product attention ($QK^\top/\sqrt{d}$) which can cause softmax saturation and unstable training. Modern architectures address this by modifying their interaction mechanism:

- **Scaled Cosine Attention:** Applies ℓ_2 normalization to queries (Q) and keys (K), projecting them onto a unit hypersphere (Liu et. al., 2022). The token interaction matrix is then scaled using a learnable temperature parameter α . This bounds interactions to $[-1, 1]$, stabilizing gradients in massive architectures while retaining the softmax operation:

$$Q'_i = \frac{Q_i}{\|Q_i\|_2} \quad K'_j = \frac{K_j}{\|K_j\|_2}$$

$$\text{Attention}(Q, K, V) = \text{softmax}(\alpha \cdot Q'K'^\top)V$$

- **Linear Cosine Attention:** In contrast, architectures like Cognition (Mongaras et. al., 2024) replace the softmax operation entirely with raw cosine similarity. Removing the non-linear softmax allows the matrix multiplications to be reordered natively, achieving true linear memory complexity with sequence length.
- **Mamba / SSM:** Token information is mixed through an input dependent recurrence:

$$h_t = \overline{A}_t h_{t-1} + \overline{B}_t x_t$$

- **Diffusion LLM:** Bidirectional scaled dot-product attention to model token interactions simultaneously across the entire sequence:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

Unlike causal autoregressive models, this non-causal interaction allows every token to look at both left and right contexts simultaneously during each iterative denoising step.

- **Mixture-of-Experts (MoE):** Scaled dot-product attention handles token-to-token interactions within the backbone. For token-to-expert interaction (routing), models rely on a linear dot-product gating mechanism. Depending on the formalism, tokens interact with gating weights (W_g) via an unconstrained inner product where the top- k selection can occur before the softmax (Shazeer et. al., 2017) or after:

$$G(x) = \text{softmax}(\text{TopK}(x \cdot W_g, k))$$

- **Comparison Status:** Different architectures have different models of token interaction.

3.3 Comparison Narratives & Methodological Differences

Narrative 1: The Measurement of Entropy is Fractured

The literature agrees that tokens compress information, but the methodology for measuring this entropy is fractured by architecture. In Transformers, tools like “Entropy-Lens” are observational—they probe the residual stream to plot a 1D entropy profile layer-by-layer (Ali et. al., 2025). In contrast, MoE literature uses entropy functionally rather than observationally; DynaMoE actively computes a token’s confidence threshold (entropy) to dynamically allocate expert compute on the fly (Gülmez, 2026). There is no direct comparative study measuring the Shannon entropy of a Mamba hidden state against a Transformer residual stream at equivalent depths.

Narrative 2: Dimensionality relies on Discrete Layers

Transformers exhibit a “Dimensional Chasm,” collapsing 10^3 parameters into ≈ 10 -dimensional semantic submanifolds (Song et. al., 2025). However, the methodological limitation here is severe: this geometric analysis heavily relies on the explicit residual stream and the discrete layer-by-layer structure of Transformers (Song et. al., 2025). Because Diffusion models iterate temporally rather than spatially, and Mamba models mediate through a continuous state space, the existing tools to measure Intrinsic Dimension simply do not port over, leaving a massive blank spot in cross-architecture geometric comparison.

3.3.1 Narrative 3: Unrolling the Black Box of Token Interaction

Historically, comparing the pairwise similarity of tokens in a Transformer to those in Mamba was impossible because Mamba squashes everything into a compressed 1D hidden state. The methodological breakthrough comes from LaTIM, which uses dynamical systems math to “unroll” Mamba’s sequential states into extractable, pairwise interaction scores, revealing that Mamba implicitly forms structured dependencies over time—prioritizing recent tokens and highly salient distant tokens (Pitorro & Treviso, 2025). This is one of the few areas where direct, apples-to-apples comparison is now methodologically possible. Complementary work shows that this implicit structure is not incidental: Mamba’s data-controlled linear operators generate “hidden” attention matrices on the fly, executing a highly factored, sequential form of self-attention (Ali et. al., 2025). Together, these findings suggest Mamba structurally mimics Transformer attention implicitly, though the two papers establish this through distinct methodologies—LaTIM through extraction of interaction scores, and Ali et. al. through direct derivation of hidden attention matrices.

Chapter 4

Literature Review of Important Papers on Different LLM architecture

This section covers the summary of the 24 most important papers of the different LLM architectures. Each summary explores several important questions, specifically:

- **Architecture(s) studied:** Which of the four families? At what scale?
- **Token-level object:** What is measured about each token? (entropy, norm, rank, position in manifold, routing decision, masking probability)
- **Analytical tool:** Information theory, dynamical systems, geometry, complexity theory, empirical probing?
- **Layer-wise vs. global:** Does the analysis track token evolution across layers, or characterize the whole model?
- **Key finding about token dynamics:** What happens to tokens as they propagate through the model?
- **Cross-architecture implications:** Does this finding generalize to other architectures, or is it architecture-specific?
- **Limitations:** What architectures/scales were not tested?

4.1 Transformer

4.1.1 Forget BIT, It is All about TOKEN

- **Source:** Bai, B. (2025). Forget BIT, it is all about TOKEN: Towards semantic information theory for LLMs. arXiv preprint arXiv:2511.01202.
- **Architecture Studied:** Transformers, Mamba, and Diffusion (LLaDA)
- **Token-level Object:** Semantic information content (entropy, directed information, rate-distortion)

- **Analytical Tool:** Information Theory
- **Layer Wise vs Global:** Global. The analysis characterizes the probabilistic model of the architectures holistically across pre-training, post-training, and inference phases.
- **Key Finding about Token Dynamics**
 Tokens (not bits) are the fundamental unit of semantic information. LLMs act as discrete-time channels, and the semantic information flow (directed information density) naturally progresses and accumulates over the token sequence.
 (Simpler Explanation) AI processes language in meaningful chunks rather than raw data. It delivers these chunks one by one, and with every new word it adds, the total "meaning" of the message grows more specific and detailed.
- **Cross-architecture Implications**
 Broadly applicable. By defining structure-agnostic information-theoretic measures, the framework is designed to evaluate Transformers, Mamba, and Diffusion models under a unified semantic umbrella.
- **Limitations**
 The framework focuses heavily on high-level mathematical probability models and global information boundaries rather than tracking the step-by-step geometric or layer-wise evolution of tokens inside the hidden states.

4.1.2 Bridging the Dimensional Chasm

- **Source:** Song, Z. Y., Li, Z., Cao, Q. H., Luo, M. X., Zhu, H. X. (2025). Bridging the dimensional chasm: Uncover layer-wise dimensional reduction in transformers through token correlation. arXiv preprint arXiv:2503.22547.
- **Architecture Studied:** Transformers
- **Token-level Object:** Position in the manifold and intrinsic dimension
- **Analytical Tool:** Geometry and empirical probing
- **Layer Wise vs Global:** Layer-wise
- **Key Finding about Token Dynamics**
 While transformers employ high dimensional embedding (10^3), effective model tend to compress tokens into approximately 10-dimensional submanifolds, closely resembling human semantic spaces.
 This work not only advances LLM interpretability by reframing Transformers layers as projectors that mediate between high-dimensional computation and low-dimensional semantics, but also provides practical tools for model diagnostics that do not rely on task-specific evaluations

- **Cross-architecture Implications**

Currently architecture-specific. As noted in your gap analysis, it remains entirely unknown if selective state spaces (Mamba) or iterative denoising processes (Diffusion) exhibit this same dimensional expansion and collapse.

- **Limitations**

The geometric analysis relies on the explicit residual stream and layer-by-layer structure of Transformers; architectures without discrete layer-wise attention mechanisms were not evaluated.

4.1.3 Attention Sinks and Compression Valleys in LLMs are Two Sides of the Same Coin

- **Source:** Queipo-de-Llano, E., Arroyo, Á., Barbero, F., Dong, X., Bronstein, M., LeCun, Y., Shwartz-Ziv, R. (2026, April). Attention sinks and compression valleys in llms are two sides of the same coin. In International Conference on Learning Representations (Vol. 2026, pp. 13911-13937).

- **Architecture Studied:** Transformers

- **Token-level Object:** Token norm (massive activations) and representational rank

- **Analytical Tool:** Empirical probing, linear algebra, and information theory

- **Layer Wise vs Global:** Layer-wise

- **Key Finding about Token Dynamics**

Proves a theoretical connection showing that massive token activations in the residual stream cause both attention sinks and compression valleys. Proposes a 3-phase "Mix-Compress-Refine" theory of how tokens flow through layers.

(Simpler Explanation)

Attention Sink is when AI directs a disproportionate amount of its "focus" (attention weights) to the very first few tokens in a sequence. Even if these initial tokens lack semantic meaning. **Compression Valley** is where intermediate representations show unexpectedly low entropy(ordered and predictable) despite the model's high-dimensional space.

Proposed 3 Phase Pattern: In the first few layers, the AI is "messy." It looks at every word in your prompt and tries to see how they all relate to each other. It's gathering all possible context.

In the middle layers, the AI gets serious. It realizes it doesn't need to keep every tiny detail in its "short-term memory." It squishes (compresses) the information into a dense, high-energy summary. This is the "Compression Valley."

Selective Refinement (The Plating): In the final layers, the AI takes that dense summary and polishes it. it stops looking at the big picture and focuses purely on which specific word should come next.

This framework provides a more mechanistic account of how LLM allocate compute across depth.

- **Cross-architecture Implications**

Highly architecture-specific. Attention sinks are a byproduct of global $O(n^2)$ self-attention and specific LayerNorm dynamics. It is unknown how representational rank evolves with depth in sequential models like SSMs or routed models like MoEs.

- **Limitations**

Phenomenon is intrinsically tied to pre-norm Transformer architectures. Models that process tokens non-autoregressively or without explicit attention matrices are outside the scope of this theory.

4.1.4 Entropy-Lens: Uncovering Decision Strategies in LLMs

- **Source:** Ali, R., Caso, F., Irwin, C., Liò, P. (2025). Entropy-lens: The information signature of transformer computations.

- **Architecture Studied:** Transformers

- **Token-level Object:** Shannon entropy of intermediate token distributions

- **Analytical Tool:** Information Theory

- **Layer Wise vs Global:** Layer-wise

- **Key Finding about Token Dynamics**

Residual Streams are simply the sum of the output of all previous layers and the original embedding. It is akin to a communication channel, since it doesn't do any processing itself and all layers communicate through it. It allows model to have multiple layers while remembering the important things that happen in each layer (like keeping a version history)

Introduces "Entropy-Lens" to map token dynamics in the residual stream to a 1D "entropy profile." Reveals that models use depth-invariant expansion and pruning strategies that are distinct to specific model families.

(Simpler Explanation) "Entropy Lens" is used to analyze AI's high dimensional and categorical dynamics of token space. It shows that LLMs use expansion(consider different possibilities) and pruning(removing unlikely words) between layers. However, each model family have different processes on how they mix expansion and pruning.

- **Cross-architecture Implications**

Highly generalizable. The primary contribution of this tool is its ability to highlight family-specific computation patterns, making it a direct comparative tool between Transformers, SSMs, and others.

- **Limitations**

While it effectively identifies what the entropy profile looks like for different families, it is primarily an observational/diagnostic tool and may not fully explain the mechanistic why behind those entropy shifts.

4.1.5 MTM: A Multi-Scale Token Mixing Transformer for Irregular Multivariate Time Series Classification

- **Source:** Zhong, S., Zhuo, W., Song, S., Li, G., Yu, Z., Chan, S. H. G. (2025, August). Mtm: A multi-scale token mixing transformer for irregular multivariate time series classification. In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2 (pp. 4074-4085).
- **Architecture Studied:** Transformers
- **Token-level Object:** Clustering behavior and relative positions in representation space.
- **Analytical Tool:** Dynamical systems (Mean-field interacting particle model)
- **Layer Wise vs Global:** Layer-wise
- **Key Finding about Token Dynamics**
When viewed as interacting particles, tokens exhibit a specific sequential evolution: a fast initial representation collapse, followed by the formation of distinct clusters, which then sequentially merge as the tokens pass through deeper layers. This paper proves the mathematical bounds of such transitions.
- **Cross-architecture Implications**
Architecture-specific. Self-attention behaves naturally like a particle interaction system. As highlighted in your survey plan, it is an open question whether sequential processing (Mamba) causes clustering, and the concept is practically non-applicable to Diffusion LLMs where tokens are intentionally corrupted and denoised rather than clustered.
- **Limitations**
The mathematical proofs heavily rely on the all-to-all dense interactions of self-attention. The model does not account for state-mediated processing, sparse routing, or diffusion trajectories.

4.1.6 The Mean-Field Dynamics of Transformers

- **Source:** Rigollet, P. (2026, July). The mean-field dynamics of transformers. In International Congress of Mathematicians 2026 (pp. 389-404). Society for Industrial and Applied Mathematics.
- **Architecture Studied:** Transformers
- **Token-level Object:** Token position, clustering, and angular dispersion on the unit hypersphere ($\mathbb{S}^{d-1}$)
- **Analytical Tool:** Geometric / Manifold and Dynamical Systems (mean-field limits and collapse theory)

- **Layer Wise vs Global:** Layer-wise. It tracks the asymptotic trajectory of tokens as layer depth approaches infinity.

- **Key Finding about Token Dynamics**

When modeling the pure self-attention mechanism mathematically, tokens are subjected to an inescapable, attractive geometric force. Because the softmax attention matrix averages values, tokens constrained to a hypersphere rapidly lose their angular dispersion. As they propagate through successive layers, this dynamic inherently drives all tokens in a sequence to collapse into a single, degenerate cluster. Essentially, **without counter-forces (like MLPs or specific residual structures), the tokens inevitably lose their distinct semantic identities and blend into an indistinguishable mass.**

Softmax attention matrix is the "weight" each token in a sequence pays to every other token. It dynamically extracts contextual information by producing a probability distribution of token relationships.

- **Cross-architecture Implications**

Architecture-specific. The proof explicitly targets the mathematical properties of the softmax self-attention matrix. As highlighted in your survey gaps, it remains unknown if the sequential processing in Mamba or iterative steps in Diffusion suffer from an analogous mathematical collapse.

- **Limitations**

This is a purely theoretical analysis of the attention mechanism in isolation. It does not account for how diverse training data, varying parameter scales, or modern architectural interventions (like MoE routing) delay or mitigate this collapse in practice.

4.1.7 You Do Not Fully Utilize Transformer's Representation Capacity

- **Source:** Gerasimov, G., Aksenov, Y., Balagansky, N., Sinii, V., Gavrilov, D. (2025). You Do Not Fully Utilize Transformer's Representation Capacity. arXiv preprint arXiv:2502.09245.

- **Architecture Studied:** Transformers

- **Token-level Object:** Routing decisions and representation separability

- **Analytical Tool:** Routing / Gating and Information-Theoretic (focusing on preserving token information capacity)

- **Layer Wise vs Global:** Layer-wise processing with global memory access.

- **Key Finding about Token Dynamics**

Standard deep Transformers force tokens into an information bottleneck; to pass context forward, a token must compress all historical data into its current layer's residual stream, which directly fuels representation collapse. By introducing a mechanism where

tokens dynamically route their queries to historical states across all previous layers (on a per-head basis), tokens are freed from this bottleneck. They no longer have to carry the burden of past context continuously, allowing their current layer representations to remain highly distinct, separable, and expressive even in extremely deep networks

- **Cross-architecture Implications**

Currently specific to Transformers, though it conceptually mirrors the "selective forgetting/remembering" goal of Mamba's input-dependent gating.

- **Limitations**

The paper focuses on resolving token collapse but does not deeply address how cross-layer routing scales computationally regarding memory bounds or extreme context lengths. **This paper suggests improvements to existing infrastructure.**

4.1.8 REFORM : Residual Filtering through Neural Aggregators for Layer-Wise Representation Integrity

- **Source:** Kang, S., Yun, W. S., Lee, N., Cho, Y. S. REFORM: Residual Filtering through Neural Aggregators for Layer-Wise Representation Integrity.

- **Architecture Studied:** Transformers

- **Token-level Object:** Token separability and hidden state uniformity

- **Analytical Tool:** Geometric/Manifold and Routing/Gating

- **Layer Wise vs Global:** Layer-wise (integrating states across depths)

- **Key Finding about Token Dynamics**

Tokens naturally suffer from severe feature homogenization (over-smoothing) as they pass through successive self-attention blocks, gradually losing their unique semantic features. REFORM introduces a hierarchical hidden state integration mechanism that acts as a structural regularization. By explicitly aggregating earlier layer outputs into current layers, it forces token trajectories to remain distinct within the manifold rather than blending into a polysemantic average. This ensures that fine-grained, token-specific information survives the journey to the final output layer.

- **Cross-architecture Implications**

While tested on Transformers to solve attention-induced smoothing, the concept of multi-layer aggregation could theoretically benefit any deep architecture prone to representational degeneration.

- **Limitations**

The study is isolated to Transformer baselines; it is untested on SSMs or MoE architectures to see if their inherent structural differences naturally resist the over-smoothing REFORM aims to fix. **This paper suggests improvements to existing infrastructure.**

4.1.9 Flowing Through Layers: A Continuous Dynamical Systems Perspective on Transformers

- **Source:** Fein-Ashley, J. (2025). Flowing through layers: A continuous dynamical systems perspective on transformers. arXiv preprint arXiv:2502.05656.
- **Architecture Studied:** Transformers
- **Token-level Object:** Continuous trajectory and perturbation decay (norm)
- **Analytical Tool:** Dynamical Systems (ODE formulation, contractive dynamics)
- **Layer Wise vs Global:** Layer-wise (but modeled as continuous time/depth)
- **Key Finding about Token Dynamics**

The discrete, step-by-step layer updates in a Transformer can be rigorously mapped to the solution of a continuous Ordinary Differential Equation (ODE). Through this lens, token dynamics are mathematically proven to be contractive. This means that as tokens move "forward in time" (deeper into the network), any initial perturbations (small, external change in input data), noise, or minor variances in their embeddings rapidly decay. While this contractive pull explains why deep Transformers are incredibly robust and resilient to input noise, it also explains their inherent vulnerability to representation collapse, as fine details are mathematically smoothed away.

Simpler explanation: As token moves through layers, they "converge" towards a single mathematical point, meaning they have lost their individual identities. By using ODEs, we can map how tokens will act downstream.

Representation Collapse: model maps diverse, distinct inputs into nearly identical internal representations. When the "contractive pull" is too strong, the tokens don't just get close; they become identical. Model is "dead" as every token in sequence is seen as the exact same concept, making it impossible to generate a coherent or nuanced response.

- **Cross-architecture Implications**
Highly cross-architectural. It establishes a direct, mathematical equivalence in fundamental token expressivity between the Transformer and Mamba families.
- **Limitations**
Circuit complexity bounds prove what an architecture can compute in theory, but they offer no guarantees about learnability—meaning it does not address whether Mamba can learn these token computations as efficiently or with the same sample complexity as a Transformer during actual training.

4.2 Architecture B: Mamba/SSM - Token Dynamics

4.2.1 The Computational Limits of State-Space Models and Mamba via the Lens of Circuit Complexity

- **Source:** Chen, Y., Li, X., Liang, Y., Shi, Z., Song, Z. (2024). The computational limits of state-space models and mamba via the lens of circuit complexity. arXiv preprint arXiv:2412.06148.
- **Architecture Studied:** Mamba/SSM
- **Token-level Object:** Computability / sequence processing capacity
- **Analytical Tool:** Computational Complexity (Circuit complexity)
- **Layer Wise vs Global:** Global. It characterizes the absolute theoretical limits of entire architecture family
- **Key Finding about Token Dynamics**

There is a radical mechanistic difference in how these architectures process tokens—Transformers use all-to-all dense interactions, while Mamba relies on sequential, state-mediated updates. However, this paper mathematically proves that both architectures fall into the exact same computational complexity class (TC^0 with polynomial precision). This means that despite Mamba tokens being processed one by one through a constrained state space, the architecture suffers no fundamental theoretical disadvantage in expressivity; it is capable of computing the exact same class of functions over a token sequence as a full self-attention mechanism
- **Cross-architecture Implications**

Highly cross-architectural. It establishes a direct, mathematical equivalence in fundamental token expressivity between the Transformer and Mamba families.
- **Limitations**

Circuit complexity bounds prove what an architecture can compute in theory, but they offer no guarantees about learnability—meaning it does not address whether Mamba can learn these token computations as efficiently or with the same sample complexity as a Transformer during actual training.

4.2.2 A Theoretical Analysis of Mamba’s Training Dynamics: Filtering Relevant Features for Generalization in State Space Models

- **Source:** Shandirasegaran, M., Li, H., Zhang, S., Wang, M., Zhang, S. (2026). A Theoretical Analysis of Mamba’s Training Dynamics: Filtering Relevant Features for Generalization in State Space Models. arXiv preprint arXiv:2602.12499.
- **Architecture Studied:** Mamba/SSM

- **Token-level Object:** Feature alignment and input-dependent gating
- **Analytical Tool:** Dynamical Systems (analyzing continuous training dynamics and sample complexity).
- **Layer Wise vs Global:** Global (focuses on how the model evolves holistically during the training process).
- **Key Finding about Token Dynamics**
Mamba’s input-dependent gating vector successfully aligns with class-relevant features while actively ignoring irrelevant ones. This proves that Mamba achieves a feature-selection role functionally similar to the ”Attention” mechanism, but realized entirely through selective recurrence
- **Cross-architecture Implications**
The findings are architecture-specific to Mamba-style selective SSMs. The paper explicitly provides a theoretical counterpoint to Transformer-centric explanations, showing how sequential models can filter tokens and guarantee generalization without all-to-all attention interactions.
- **Limitations**
First, the theoretical analysis focuses on a simplified Mamba setting that abstracts away practical components such as depth, multiple heads, residual connections, and layer normalization. Second, the data model, while standard in theoretical studies, also simplifies real-world sequence structures. Extending the analysis to more realistic multi-layer and multi-head Mamba architectures, richer data models, and alternative designs such as gated Transformers or hybrid Mamba-Transformer frameworks remains absent from current work.

4.2.3 Understanding Input Selectivity in Mamba: Impact on Approximation Power, Memorization, and Associative Recall Capacity

- **Source:** Huang, N., Sarabia, M., Moudgil, A., Rodriguez, P., Zappella, L., Danieli, F. (2025). Understanding input selectivity in mamba: impact on approximation power, memorization, and associative recall capacity. arXiv preprint arXiv:2506.11891.
- **Architecture Studied:** Mamba/SSM
- **Token-level Object:** Token memory retention and information decay
- **Analytical Tool:** Routing / Gating and Information-Theoretic (memory constraints)
- **Layer Wise vs Global:** Layer-wise
- **Key Finding about Token Dynamics**
Standard recurrent networks and basic SSMs suffer from exponential memory decay; old tokens simply fade away. This paper mathematically maps Mamba’s core selective

mechanism (the S6 layer) to Haar wavelet projections. By dynamically adjusting these wavelet projections on a per-token basis, Mamba actively counteracts that natural exponential decay. It acts as a sophisticated, multi-scale filter that can deliberately hold onto specific, highly relevant historical tokens for massive context windows, while aggressively forgetting useless intermediary tokens

- **Cross-architecture Implications**

It highlights how Mamba achieves "routing-like" temporal selectivity, which is the sequential equivalent of a Transformer's explicit ability to look back at any past token via attention.

- **Limitations**

While the Haar wavelet projection is a brilliant theoretical mapping, it is a mathematical abstraction that may not perfectly capture the chaotic, non-linear interactions of a fully trained, multi-layer Mamba model in practice.

4.2.4 LaTIM: Measuring Latent Token-to-Token Interactions in Mamba Models

- **Source:** Pitorro, H., Treviso, M. V. (2025, July). Latim: Measuring latent token-to-token interactions in mamba models. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 24478-24493).

- **Architecture Studied:** Mamba/SSM

- **Token-level Object:** Pairwise interaction strength between tokens

- **Analytical Tool:** Information-Theoretic and Dynamical Systems (decomposing sequential states).

- **Layer Wise vs Global:** Layer-wise

- **Key Finding about Token Dynamics**

Mamba processes tokens sequentially by squashing them into a compressed 1D hidden state, seemingly destroying the direct token-to-token maps seen in Transformers. However, LaTIM proves you can mathematically "unroll" this sequential process to extract exact, fine-grained pairwise interaction scores. The key finding is that Mamba implicitly forms highly structured dependencies over time—dynamically prioritizing recent tokens and highly salient distant tokens—without ever actually computing a full $O(n^2)$ matrix.

- **Cross-architecture Implications**

This provides the exact mathematical tools needed to extract an "attention map" from an SSM, allowing for direct, apples-to-apples comparisons between Mamba's sequential interactions and a Transformer's global attention.

- **Limitations**

Reconstructing these pairwise interactions from a compressed state is an intensive analytical exercise; it is an observational tool for researchers rather than a lightweight process that can be used dynamically during the model’s forward pass.

4.2.5 The Hidden Attention of Mamba Models

- **Source:** Ali, A. A., Zimmerman, I., Wolf, L. (2025, July). The hidden attention of mamba models. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 1516-1534).

- **Architecture Studied:** Mamba/SSM

- **Token-level Object:** Implicit attention weights instantiated by the token stream

- **Analytical Tool:** Routing/Gating and Dynamical Systems

- **Layer Wise vs Global:** Layer-wise

- **Key Finding about Token Dynamics**

Mamba does not merely approximate attention. It generates ”hidden” attention matrices on the fly. The data-controlled linear operators in Mamba act as micro-routers.

Simpler Explanation: Mamba belongs to a family of AI called ”Selective State-Space Models.” Older state-space models applied the exact same math to every piece of incoming data. Mamba is ”selective”—its mathematical filters (the linear operators) change dynamically based on the specific word it is looking at right now (data-controlled). This allows Mamba to get the primary benefit of the attention mechanism(contextual information) without having to calculate the giant grid. The attention is ”hidden in the math of the state space model.

The paper reveals a staggering fact: a standard Mamba model effectively utilizes three orders of magnitude more hidden attention matrices than a comparably sized Transformer has explicit attention heads. Instead of a few global heads looking at everything, Mamba processes tokens through a massive, distributed ensemble of highly localized, implicit attention routers. Thus, by replacing a few giant, slow attention mechanisms with a massive army of highly efficient, dynamic ”micro-routers,” Mamba can process information incredibly fast without losing the ability to understand deep context.

- **Cross-architecture Implications**

This completely blurs the architectural lines. It suggests that data-dependent SSMs and Transformers are not as functionally different as their code implies; Mamba is simply executing a highly factored, sequential form of self-attention.

- **Limitations**

”Hidden attention” is an implicit mathematical property. Because Mamba cannot physically jump back to explicitly retrieve a raw token from the past like a Transformer, it still struggles with exact retrieval tasks (like the needle-in-a-haystack problem) despite these hidden matrices.

4.3 Diffusion LLM — Token Dynamics

4.3.1 Large Language Diffusion Models

- **Source:** Nie, S., Zhu, F., You, Z., Zhang, X., Ou, J., Hu, J., ... Li, C. (2026). Large language diffusion models. *Advances in Neural Information Processing Systems*, 38, 50608-50646.
- **Architecture Studied:** Diffusion LLM
- **Token-level Object:** Masking probability and bidirectional context confidence
- **Analytical Tool:** Information-Theoretic (Iterative denoising and forward/reverse probabilities).
- **Layer Wise vs Global:** Global (However, instead of analyzing spatial layers, the analysis tracks tokens across temporal denoising timesteps)
- **Key Finding about Token Dynamics**

Tokens in LLaDA completely break the standard autoregressive (left-to-right) lifecycle. Instead of accumulating meaning layer by layer, tokens are processed through forward masking (intentional corruption) and reverse prediction (iterative denoising). During inference, all tokens are updated simultaneously. The model continuously refines its confidence across the entire sequence bidirectionally, updating the probability distribution of each individual token based on the gradually emerging global context. This allows it to achieve competitive reasoning performance with strong autoregressive models like LLaMA3-8B, but via a radically different mechanism. For demo, view : <https://ml-gsai.github.io/LLaDA-demo/>
- **Cross-architecture Implications**

It implies a different perspective on how LLM are derived.
- **Limitations**

Because the architecture is so new, the deep mechanistic dynamics of the tokens are still largely a black box. It is currently unknown whether tokens in a Diffusion LLM cluster, collapse, or traverse low-dimensional submanifolds the way Transformer tokens do during these denoising steps.

4.3.2 Simple and Effective Masked Diffusion Language Models

- **Source:** Sahoo, S. S., Arriola, M., Schiff, Y., Gokaslan, A., Marroquin, E., Chiu, J. T., ... Kuleshov, V. (2024). Simple and effective masked diffusion language models. *Advances in Neural Information Processing Systems*, 37, 130136-130184.
- **Architecture Studied:** Diffusion LLM
- **Token-level Object:** Token masking probability and ELBO(Evidence Lower Bound)

- **Analytical Tool:** Information Theory(ELBO optimization, cross-entropy bounds)
- **Layer Wise vs Global:** Global (tracking tokens across temporal denoising steps)

- **Key Finding about Token Dynamics**

Traditional autoregressive models learn token distributions by explicitly calculating exact next-token probabilities. Diffusion models, however, learn by reversing a noise-corruption process. A major challenge in discrete token spaces is that calculating the true reverse probability involves high-variance Monte Carlo estimates over the entire vocabulary. MDLM reveals that the token-level learning objective can be mathematically reformulated using Rao-Blackwellization—a statistical technique that reduces variance by conditioning on a subset of variables. Technically, this rewrites the complex diffusion objective into a mixture of standard masked cross-entropy losses. Instead of guessing blindly, the model computes a marginalized expectation over the forward noise process. This leads to a provably tighter Evidence Lower Bound (ELBO), meaning the model captures the true underlying token information density much more efficiently. As tokens undergo the reverse process, their semantic certainty increases continuously, effectively proving that masked diffusion is mathematically homologous to a continuous, iterative version of Masked Language Modeling (like BERT).

- **Cross-architecture Implications**

This is specific to masked diffusion objectives, though it conceptually bridges discrete diffusion math with the standard optimization objectives used in encoder-only Transformers

- **Limitations**

The paper strictly focuses on optimization objectives and probabilistic bounds. It does not explain the spatial interactions—meaning it doesn’t analyze how Token A geometrically influences Token B within the hidden layers during a specific denoising step.

4.3.3 Balancing Understanding and Generation in Discrete Diffusion Models

- **Source:** Liu, Y., Zhao, Y., Xie, Z., Ye, Q., Jiao, J., Hu, Y., ... Liu, Y. (2026). Balancing understanding and generation in discrete diffusion models. arXiv preprint arXiv:2602.01362.
- **Architecture Studied:** Diffusion LLM
- **Token-level Object:** Transition probability matrices and noise kernels
- **Analytical Tool:** Information-Theoretic and Dynamical Systems (Markov transition dynamics)
- **Layer Wise vs Global:** Global (temporal trajectory)

- **Key Finding about Token Dynamics**

In discrete language diffusion, tokens are usually corrupted in one of two distinct ways: Masked diffusion (replacing a token entirely with a [MASK] token) or Uniform diffusion (randomly flipping a token to another word in the vocabulary). XDLM discovers a theoretical stationary noise kernel that mathematically unifies both paradigms. By defining token corruption as a unified Markov transition matrix, the research shows that tokens transition through a predictable probability space where masking acts as an "absorbing state" and uniform flipping acts as an "ergodic mixing process." The key technical breakthrough is proving that the reverse-time dynamics (the actual generation process) can smoothly interpolate between these two extremes. Tokens can initially be retrieved from uniform noise (which establishes broad semantic and contextual domains) and later refined through mask-prediction mechanics (which locks in exact syntax and grammar).

- **Cross-architecture Implications**

This uniquely solves a theoretical division within the Diffusion LLM family itself, creating a unified mathematical framework for discrete noise. It does not apply to autoregressive standard Transformers or SSMs.

- **Limitations**

The mathematical proofs assume a fixed, distinct vocabulary space. It does not account for sub-word tokenization geometries (e.g., how the semantic distance between specific BPE subwords affects the optimal uniform transition matrix).

4.3.4 Inference-Time Scaling of Diffusion Language Models via Trajectory Refinement

- **Source:** Dang, M., Han, J., Xu, M., Xu, K., Srivastava, A., Ermon, S. (2025). Inference-Time Scaling of Diffusion Language Models via Trajectory Refinement. arXiv preprint arXiv:2507.08390.

- **Architecture Studied:** Diffusion LLM

- **Token-level Object:** Generation trajectory and posterior token distribution

- **Analytical Tool:** Dynamical Systems (Markov Chain Monte Carlo, Particle Gibbs sampling)

- **Layer Wise vs Global:** Global(trajecotry-level)

- **Key Finding about Token Dynamics**

A major mechanical flaw in standard diffusion language models is the accumulation of conditioning errors: if a token is incorrectly predicted early in the reverse-denoising path, that error severely corrupts the conditioning context for all surrounding tokens, leading to a degraded final sequence. PG-DLM models the entire sequence generation as a dynamic state-space trajectory. By applying Particle Gibbs (a sequential Monte Carlo method), the model runs multiple parallel token trajectories ("particles") and

dynamically resamples them based on their likelihood scores at intermediate denoising steps. The technical achievement is providing strict theoretical convergence guarantees that this resampling actively pulls the token distribution back toward the true data manifold. Essentially, tokens are allowed to mathematically "change their minds" mid-generation if their current trajectory leads to low-probability semantic states, actively healing context errors before the final text is output

Particle Gibbs sampling algorithm is an advanced Markov Chain Monte Carlo method that uses Sequential Monte Carlo or particle filtering to estimate complex probability distributions

- **Cross-architecture Implications**

While the continuous math is applied to Diffusion, the use of trajectory-level resampling conceptually mirrors search algorithms like best-first tree search or beam search used during Transformer inference.

- **Limitations**

Particle filtering is notoriously expensive computationally. The theoretical gains in token accuracy come at a massive latency cost during inference, making it difficult to scale to real-time text generation.

4.4 Mixture-of-Experts: Token Routing

4.4.1 DynaMoE: Dynamic Token-Level Expert Activation with Layer-Wise Adaptive Capacity for Mixture-of-Experts Neural Networks

- **Source:** Gülmez, G. (2026). DynaMoE: Dynamic Token-Level Expert Activation with Layer-Wise Adaptive Capacity for Mixture-of-Experts Neural Networks. arXiv preprint arXiv:2603.01697.

- **Architecture Studied:** Mixture-of-Experts

- **Token-level Object:** Routing decision, expert activation count, and compute load per token.

- **Analytical Tool:** Routing/Gating

- **Layer Wise vs Global:** Layer-wise

- **Key Finding about Token Dynamics**

Standard MoE architectures enforce a rigid Top- K routing policy (e.g., routing every token to exactly 2 experts), which mechanically assumes every token requires the exact same amount of computational work to process. DynaMoE overturns this by analyzing the intrinsic "difficulty" of individual tokens. The paper implements a gating mechanism that dynamically allocates a variable number of experts based on a token's computed confidence threshold. Technically, simple syntactic tokens (like "the,"

”and,” or punctuation) might be routed to a single expert or even bypass the MoE layer entirely (0 experts), while highly complex semantic tokens (like rare nouns, foreign words, or mathematical operators) trigger 4 or 5 experts simultaneously. This proves that token representations inherently contain a measurable ”complexity gradient.” By utilizing layer-wise adaptive capacity scheduling, the model prevents expert load imbalance while saving massive amounts of FLOPs on ”easy” tokens, explicitly linking a token’s information entropy to its computational routing requirement.

- **Cross-architecture Implications**

Highly MoE-specific. However, the underlying philosophy mirrors early-exit or variable-compute mechanisms sometimes grafted onto standard Transformers to save inference costs.

- **Limitations**

The paper leaves open a critical geometric question: if tokens use vastly different computational paths (some skipping layers, some using 5 experts), do their geometric positions in the final residual stream still align harmoniously, or does this dynamic routing lead to representation fragmentation?

4.4.2 Grassmannian Mixture-of-Experts: Concentration-Controlled Routing on Subspace Manifolds

- **Source:** Shihab, I. F., Akter, S., Sharma, A. (2026). Grassmannian Mixture-of-Experts: Concentration-Controlled Routing on Subspace Manifolds. arXiv preprint arXiv:2602.17798.

- **Architecture Studied:** Mixture-of-Experts

- **Token-level Object:** Token masking probability and ELBO (Evidence Lower Bound) contributions

- **Analytical Tool:** Geometric/Manifold(ELBO optimization, cross-entropy bounds)

- **Layer Wise vs Global:** Global (tracking tokens across temporal denoising steps)

- **Key Finding about Token Dynamics**

Traditional MoE routing uses a simple linear projection followed by a softmax to assign tokens to experts. This flat Euclidean approach suffers heavily from ”expert collapse” (where all tokens flood the same few experts) and noisy, unstable routing decisions during training. This paper reformulates routing as a purely geometric problem by mapping token representations and expert centroids onto a Grassmann manifold (a curved mathematical space parameterized by linear subspaces rather than points). Instead of standard softmax logits, tokens are routed based on the Bingham distribution, which naturally models antipodally symmetric probabilities on hyperspheres and Grassmannians. Technically, this geometric shift proves that expert collapse is largely an artifact of flat routing spaces. In the curved Grassmannian geometry, experts naturally repel each other into orthogonal subspaces. The paper provides formal mathematical

bounds proving that under this geometry, routing entropy is perfectly preserved and expert collapse is mathematically impossible.

- **Cross-architecture Implications**

While strictly analyzing MoE gating functions, it introduces powerful manifold-level analysis that could be used to study how dimensional reduction (like that seen in standard Transformers) impacts routing efficiency in hybrid models.

- **Limitations**

Constraining the routing function to a Grassmannian manifold introduces complex Riemannian optimization during backpropagation. The paper primarily focuses on formal theoretical bounds rather than validating training stability at frontier scales (e.g., 70B+ parameters).

4.4.3 Mixture-of-Transformers (MoT)

- **Source:** Liang, W., Yu, L., Luo, L., Iyer, S., Dong, N., Zhou, C., ... Lin, X. V. (2024). Mixture-of-transformers: A sparse and scalable architecture for multi-modal foundation models. arXiv preprint arXiv:2411.04996.

- **Architecture Studied:** Mixture-of-Experts

- **Token-level Object:** Routing decision (expert assignment) and gradient geometry.

- **Analytical Tool:** Routing / Gating and Geometric / Manifold (focusing on subtask strong convexity).

- **Layer Wise vs Global:** Global (focuses on training dynamics and convergence).

- **Key Finding about Token Dynamics**

This paper introduces a new MoE model that is designed to address the computational challenges of multi-modal model pretraining.

Traditional dense models process all tokens through the same parameters, which often causes geometric "gradient conflicts" where updating weights to improve one token type degrades the optimization path for another. By routing tokens to specialized Transformer experts, the global, highly non-convex loss landscape is partitioned into distinct, strongly convex subtasks. This token specialization ensures that gradients from different semantic token clusters are orthogonal and do not interfere with one another, enabling an exponentially fast $O(\log(\varepsilon^{-1}))$ convergence rate for specialized semantic subsets.

- **Cross-architecture Implications**

The principle that token-level specialization resolves gradient conflicts is conceptually relevant to Hybrid models that selectively route tokens to fundamentally different underlying mechanisms.

- **Limitations**

The analysis heavily prioritizes convergence rates and optimization geometry; it does

not detail how the intrinsic dimension or the Shannon entropy of the tokens changes sequentially as a result of passing through these experts.

4.4.4 DiffMoE: Dynamic Token Selection for Scalable Diffusion Transformers

- **Source:** Shi, M., Yuan, Z., Yang, H., Wang, X., Zheng, M., Tao, X., ... Gai, K. (2025). Diffmoe: Dynamic token selection for scalable diffusion transformers. arXiv preprint arXiv:2503.14487.
- **Architecture Studied:** Mixture-of-Experts + Diffusion LLMs (Diffusion Transformers)
- **Token-level Object:** Routing decision across temporal denoising steps.
- **Analytical Tool:** Routing / Gating and Dynamical Systems
- **Layer Wise vs Global:** Global (tracks tokens temporally across the iterative generation process, rather than just spatially across layers)
- **Key Finding about Token Dynamics**
: In standard language diffusion, tokens are processed uniformly at every denoising timestep, which is wasteful because tokens resolve at different speeds. DiffMoE introduces a dynamic, token-level routing mechanism specifically adapted for this continuous dynamical system. Instead of independent layer-wise routing, it implements batch-level global token pools that continuously assess a token’s confidence state. Tokens that achieve high certainty early in the reverse process are dynamically routed to "lighter" experts or dropped entirely, while highly uncertain tokens are routed to high-capacity experts. This proves that a token’s computational requirement drastically decays as it moves along the diffusion trajectory.
- **Cross-architecture Implications**
This uniquely crosses the boundary between Diffusion and MoE, proving that MoE routing principles can be mapped onto the temporal axis of iterative denoising, not just the spatial depth of an autoregressive model.
- **Limitations**
The paper focuses primarily on computational load balancing and does not deeply analyze the information-theoretic bounds of how dropping tokens affects the strict mathematical guarantees of the diffusion reverse process.

4.5 Hybrid Architectures

4.5.1 Nemotron 3 Super: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning

- **Source:** Chandiramani, A., Blakeman, A., Olaoye, A., Gupta, A., Somasamudramath, A., Khattar, A., ... Briones, E. (2026). Nemotron 3 super: Open, efficient mixture-of-experts hybrid mamba-transformer model for agentic reasoning. arXiv preprint arXiv:2604.12374.
- **Architecture Studied:** Hybrid (Mamba + Transformer + MoE)
- **Token-level Object:** Latent routing decisions
- **Analytical Tool:** Routing/Gating
- **Layer Wise vs Global:** Layer-wise (interleaved processing)
- **Key Finding about Token Dynamics**
Enforcing rigid boundaries between dense sequence processing and sequential processing limits token expressivity. This architecture implements "latent MoE," where the routing space is fundamentally expanded into a continuous latent space that seamlessly interacts with both Mamba state spaces and Transformer attention heads. The hybrid routing mechanism acts as an intelligent arbiter: tokens requiring long-context historical lookups are dynamically processed by sequential Mamba layers, while tokens requiring exact associative recall trigger dense self-attention layers. This dynamically optimizes sequence processing based on the specific semantic memory demands of each individual token.
- **Cross-architecture Implications**
Highly applicable to frontier models, demonstrating how the fundamental token processing mechanisms of Transformers, SSMs, and MoEs can be fused into a single unified routing framework.
- **Limitations**
As an applied frontier model paper, it relies heavily on empirical scaling results and lacks the rigorous mathematical proofs (like circuit complexity bounds or strict ODE formulations) explaining the fundamental theory of why tokens self-organize this way.

4.5.2 Jamba-1.5: Hybrid Transformer-Mamba Models at Scale

- **Source:** Lieber, O., Lenz, B., Bata, H., Cohen, G., Osin, J., Dalmedigos, I., ... Shoham, Y. (2024). Jamba: A hybrid transformer-mamba language model. arXiv preprint arXiv:2403.19887.
- **Architecture Studied:** Hybrid (interleaved Mamba + Transformer + MoE layers)
- **Token-level Object:** Memory retention and computational pathway assignment

- **Analytical Tool:** Routing / Gating and Information-Theoretic
- **Layer Wise vs Global:** Layer-wise (tracking tokens as they pass through heterogeneous blocks)
- **Key Finding about Token Dynamics**
By interleaving Mamba layers with Transformer layers, the token sequence is subjected to an alternating computation rhythm. During Mamba layers, tokens aggressively filter information by compressing historical context into a fixed-size hidden state. To counter the exact-retrieval limitations of this compressed state, tokens are periodically passed through explicit self-attention layers. The Mamba layers act as a highly efficient background context synthesizer, while the intermittent Transformer layers act as sharp, precise memory retrieval mechanisms. This alternating flow proves that tokens do not need to carry exact pairwise attention scores at every layer; computing dense attention sparsely is sufficient to maintain global sequence coherence and bypass the information bottleneck of pure state spaces
- **Cross-architecture Implications**
Validates the hypothesis that sequential state-mediated processing and dense all-to-all processing are complementary, rather than mutually exclusive, ways to evolve token representations
- **Limitations**
The paper does not provide a granular geometric analysis of how a token’s intrinsic dimension or manifold shape expands and contracts as it abruptly shifts from a Mamba block into a Transformer block

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